MH3510 REGRESSION ANALYSIS

Modelling Annual Average Daily Traffic

Multiple linear regression with R

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Overview

This study models traffic volume for 121 road sections using county population, lane count, road width and access control. The initial model is checked using residual plots and revised through transformations. All 11 candidate models, including unsuccessful attempts, are retained.

A fifth-root response model with log population, log lane count and access control provides a defensible final specification. For the requested road, its back-transformed typical prediction is approximately **5,494 vehicles/day**, with an approximate 95% individual prediction interval of **1,164 to 17,928 vehicles/day**. The main report follows the five-stage modelling workflow. Appendices contain diagnostic evidence, the complete R script and its output.

1 Step 1: Data and graphical exploration

Average annual daily traffic (AADT) is the average number of vehicles passing through a road section each day over a year. The analysis uses only the first five columns of `aadt.txt`; the steam dataset is not used.

Column	Symbol	Definition
1	Y	AADT, in vehicles/day (response)
2	X_1	Population of the county containing the road section
3	X_2	Number of lanes
4	X_3	Road width, in feet
5	A	Access indicator: 1 for access control; 0 for no access control

The original fifth column uses 1 for access control and 2 for no control. Set $A = 1$ for original code 1 and $A = 0$ otherwise. All 121 observations are complete: 27 have access control and 94 do not. AADT ranges from 201 to 155,547.

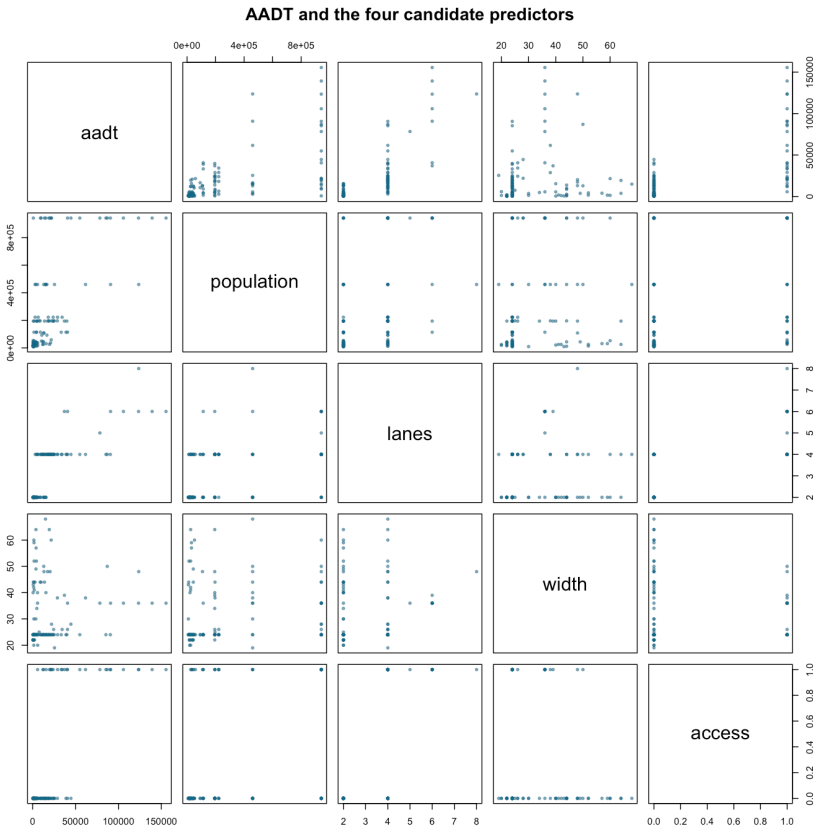


Figure 1: Scatterplot matrix of the response and four candidate predictors.

Traffic has positive marginal associations with population, lanes and access control, while width has a weaker association. Population and AADT are strongly skewed. Lanes and access control are related: all two-lane sections lack access control, while sections with more than four lanes have access control. The regression estimates conditional associations, not causal effects.

2 Step 2: Initial multiple linear regression

The starting model includes all four required predictors:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 A_i + \varepsilon_i.$$

Ordinary least squares gives

$$\widehat{Y} = -26035.84 + 0.033032 X_1 + 9157.94 X_2 + 100.29 X_3 + 23610.36 A.$$

The initial model has $R^2 = 0.7527$, adjusted $R^2 = 0.7442$ and residual standard error 15,292.6. Its overall F-test is significant ($F(4, 116) = 88.29$, $p < 2.2 \times 10^{-16}$).

Why this fit is insufficient

The fitted relationship explains some variation, but statistical significance does not establish adequacy. The residual-versus-fitted plot shows curvature and increasing spread; the Q–Q plot has substantial tail departures. Moreover, 45 fitted AADT values are negative, although traffic counts cannot be negative. These features motivate transformations rather than accepting the initial fit.

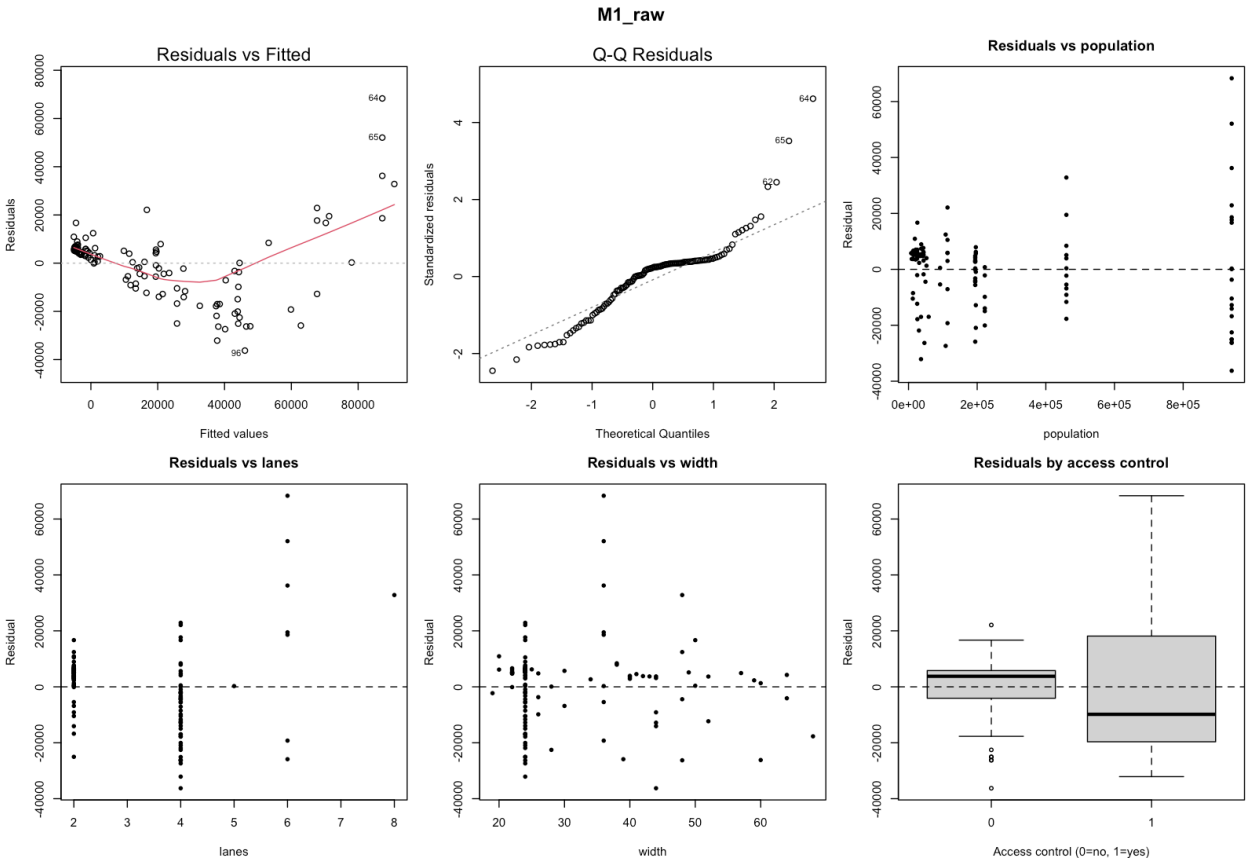


Figure 2: Diagnostics for M1, the untransformed four-predictor model. Residuals are plotted against fitted values and every predictor; the Q–Q plot checks approximate normality.

The remedies follow Chapter 2, Part 4: transform predictors to address curvature, transform the response to address changing spread, and refit and check again [2]. The fitted coefficients and full console output are retained in Appendix C.

3 Step 3: Transformations and the model record

The candidate sequence is exploratory. Logarithms compress high values strongly; the square-root transformation compresses them less. Fourth and fifth roots are intermediate positive-power trials. A fifth root is simply $Y^{1/5}$; fitting it still uses ordinary `lm()` regression, linear in its coefficients.

The predictor sets used below are $P_0 = (X_1, X_2, X_3, A)$, $P_1 = (\log X_1, X_2, X_3, A)$, and $P_2 = (\log X_1, \log X_2, X_3, A)$. A reduced set omits width X_3. All logarithms are natural.

Model	Response; predictors	What improved or remained problematic
M1	Y; P_0	Baseline: curvature, increasing spread, tail problems and negative fitted values.
M2	$\log Y$; P_0	Q–Q plot improves, but the residual trend curves downward at large fitted values.
M3	$\log Y$; P_1	Population transformation helps; curvature, uneven spread and tail observations remain.
M4	$\log Y$; P_2	Flatter mean trend, but residual spread decreases as fitted values increase.
M5	$\log Y$; P_1 plus X_2^2	Quadratic lanes improve M3, but do not resolve the changing spread.
M6	$\log Y$; reduced P_2	Width can be removed; the variance issue persists.
M7	$Y^{1/4}$; P_2	Much more stable spread and straighter Q–Q plot. A reasonable candidate.
M8	$\sqrt{Y}$; P_2	Some improvement, but curvature and increasing spread remain.
M9	$Y^{1/4}$; reduced P_2	Reasonable simpler fourth-root alternative; width is not supported.
M10	$Y^{1/5}$; P_2	Relatively flat residual trend, more stable spread and reasonable Q–Q plot.
M11	$Y^{1/5}$; reduced P_2	Selected working model: M10 simplified without losing a supported predictor.

Table 1: All candidate models are retained, including unsuccessful fits.

How the response power was chosen

The main diagnostic justification is that both root models correct the pronounced defects of the raw, log and square-root fits. M11 is a defensible working choice, not a uniquely proven optimum; M9 remains a reasonable alternative.

For transparency, earlier exploratory Box–Cox profiling suggested a power near 0.18 and led to the nearby fifth-root trial. That extended work is archived. It is not required in the course-level script or used here as a substitute for the residual checks. The exact fifth root is an extension of the lecture's transformation principle, permitted by the assignment, rather than a transformation prescribed in the slides.

Comparison rule. R^2, adjusted R^2 and residual standard deviations refer to the fitted response scale. They should not be ranked across different response transformations as if the response were unchanged. An F-test also cannot compare different transformed responses.

4 Step 4: Simplification and the final fitted model

For nested models with the same response, the extra-sum-of-squares test is

$$F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_F)/(p_F - p_R)}{\mathrm{SSE}_F/(n - p_F - 1)}.$$

Here p_F and p_R count predictors excluding the intercept. This is the full/reduced-model procedure in Chapter 4 Part II [4].

Nested comparison	F	p-value
Add width to reduced log model: M6 vs M4	0.395	0.531
Add quadratic lanes: M3 vs M5	9.165	0.00305
Add width to fourth-root model: M9 vs M7	0.723	0.397
Add width to fifth-root model: M11 vs M10	0.632	0.428
Add squared log predictors to M11 jointly	0.736	0.481

Table 2: Model comparisons. Preliminary log-model tests are interpreted alongside their unsatisfactory residual diagnostics.

At the 5% level, there is insufficient evidence that width adds explanatory value to the fifth-root model after accounting for population, lanes and access control. The residual plot against the omitted width also shows no obvious remaining trend. The additional curvature test does not support the two extra squared terms.

Let $Z = Y^{1/5}$. The selected model is

$$Z_i = \beta_0 + \beta_1 \log X_{1i} + \beta_2 \log X_{2i} + \beta_4 A_i + \varepsilon_i,$$

with fitted conditional mean

$$\boxed{\widehat{Z} = -2.739015 + 0.535189 \log X_1 + 2.317245 \log X_2 + 1.106497 A.}$$

Term	Estimate	Std. error	t	p-value
Intercept	-2.739015	0.554451	-4.940	2.63×10^{-6}
$\log X_1$	0.535189	0.052970	10.104	1.24×10^{-17}
$\log X_2$	2.317245	0.246455	9.402	5.58×10^{-16}
Access A	1.106497	0.213759	5.176	9.51×10^{-7}

Table 3: Final model coefficient estimates and individual t-tests.

On the fifth-root scale, $R^2 = 0.8528$, adjusted $R^2 = 0.8490$ and $s = 0.7478$ with 117 residual degrees of freedom. The overall test gives $F(3, 117) = 225.95$, $p < 2.2 \times 10^{-16}$. All retained predictors have evidence of a nonzero conditional association.

Final residual checks and interpretation

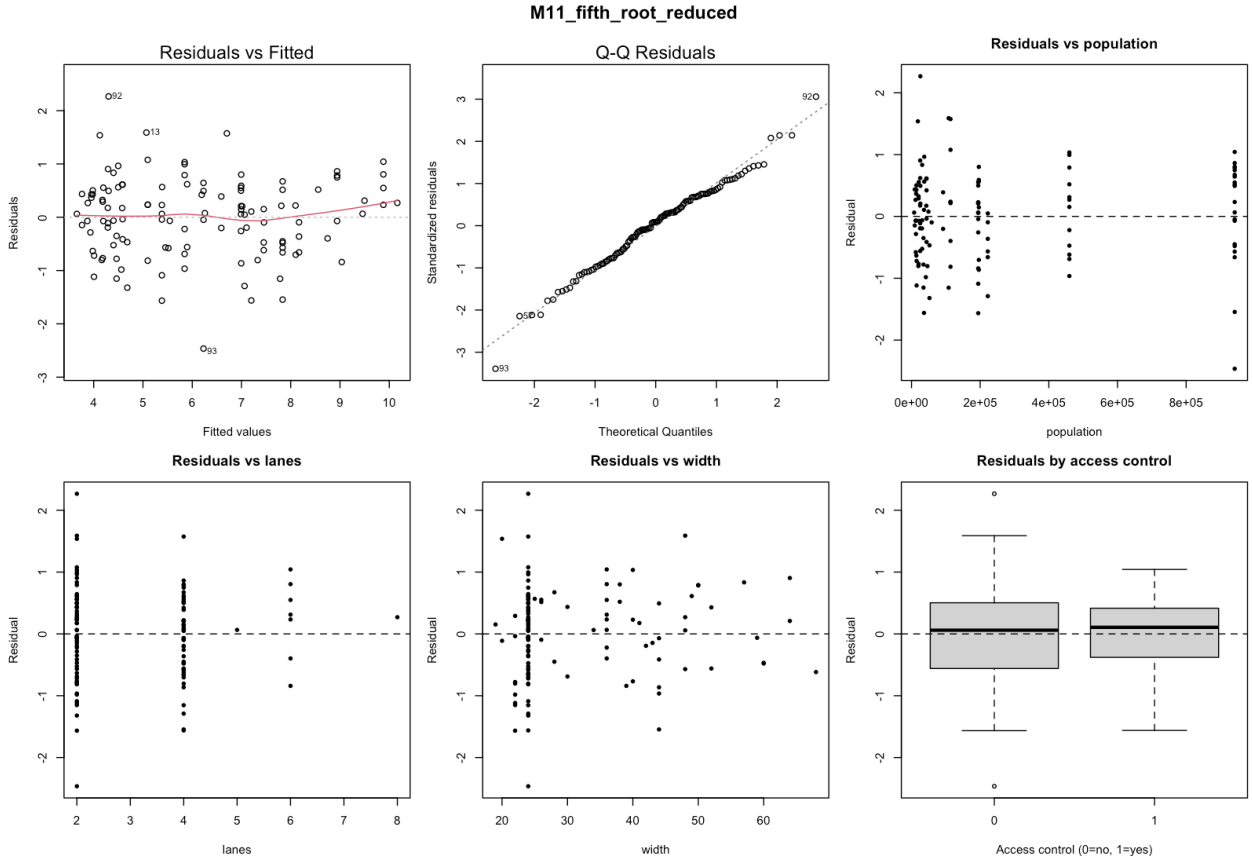


Figure 3: Final-model diagnostics. Residual spread is substantially improved, with a relatively flat fitted-value trend. Tail observations remain and must be discussed.

The residual cloud is substantially more satisfactory than M1's, and the Q–Q plot is reasonably straight apart from a few tail observations. The plots support an approximate working model; they do not prove the assumptions.

Holding the other fitted predictors fixed, doubling population increases fitted Z by $0.535189 \log 2 \approx 0.371$; doubling lane count increases it by $2.317245 \log 2 \approx 1.606$. Access control is associated with a 1.1065 higher fitted Z. These are fifth-root-scale changes, not fixed changes in vehicles/day or percentages of AADT. The associations do not establish causal effects.

Influence. Row 93 has the largest Cook's distance: AADT 756, population 941,411, two lanes, width 24 feet and no access control. Row 92 also has a large positive residual. With no evidence of recording errors, all observations remain. Omitting row 93 only for sensitivity changes the requested typical prediction from 5,494 to 5,430, about 1.2%; retained coefficient signs remain positive.

Independence. The file-order Durbin–Watson statistic is 1.7726. However, no time or spatial ordering is documented, so it does not justify a serial-correlation conclusion. Independence remains an assumption; possible county clustering cannot be resolved from the supplied variable description. The influence and row-order plots are retained in Appendix A.

5 Step 5: Prediction for the requested road

The requested setting is population 50,000, three lanes, width 60 feet and original control code 2. Since code 2 means no access control, use $A = 0$. Width is recorded but does not enter the final reduced model.

```
new_road <- data.frame(population = 50000, lanes = 3,
                       width = 60, access = 0)
predict(final_model, new_road, interval = "confidence")
predict(final_model, new_road, interval = "prediction")
```

With $\mathbf{x}_0 = (1, \log 50000, \log 3, 0)^\mathsf{T}$ and design matrix X, the lecture's intervals on the fitted Z scale are

$$\text{Mean-response CI:} \quad \widehat{Z}_0 \pm t_{0.975,117} s \sqrt{\mathbf{x}_0^\mathsf{T}(X^\mathsf{T}X)^{-1}\mathbf{x}_0},$$
$$\text{Individual PI:} \quad \widehat{Z}_0 \pm t_{0.975,117} s \sqrt{1 + \mathbf{x}_0^\mathsf{T}(X^\mathsf{T}X)^{-1}\mathbf{x}_0}.$$

The extra 1 accounts for the variability of an individual road section.

Quantity	Fitted value	Lower 95%	Upper 95%
Mean fifth-root AADT: confidence interval	5.59736	5.40290	5.79183
Individual fifth-root AADT: prediction interval	5.59736	4.10375	7.09098
Typical AADT and individual prediction interval	5494.36	1163.86	17927.96

Table 4: Prediction results. The last row is in vehicles/day; the first two rows are in $(\text{vehicles/day})^{1/5}$.

The final row raises the fitted value and individual interval endpoints to the fifth power. Because this transformation is increasing, the individual interval can be back-transformed in this way.

Mean versus typical traffic. The value $5.59736^5 \approx 5494$ is a back-transformed typical prediction (approximately a conditional median under the normal-error working model), not the arithmetic mean AADT. Similarly, raising the mean-Z confidence limits to the fifth power does not give a confidence interval for arithmetic mean AADT. The mean confidence interval is reported on its fitted scale, without claiming an original-scale mean interval or introducing advanced retransformation methods.

No three-lane sections are observed, so the requested lane count interpolates between two and four lanes. The wide individual prediction interval reflects substantial road-to-road variation and relies on the model assumptions.

6 Conclusion and limitations

A transformed multiple regression addresses the strongest deficiencies of the initial model. M11 retains population, lanes and access control; width is omitted using the full/reduced F-test and residual checks. M9 is a reasonable alternative, giving a typical prediction near 5,728. The analysis is exploratory and uses the same dataset to select and assess the model. The reported textbook intervals are conditional on the selected specification and approximate error assumptions.

Reproducibility and references

The analysis uses base R and all 121 observations. From the project directory run:

```
Rscript analysis.R
```

This generates the complete output, model objects and plots in `coursework_results/`. The supplied LaTeX source package includes the script, dataset, console output and plot assets. To compile the report from that package's directory:

```
tectonic aadt_report.tex
```

Appendix A preserves the remaining diagnostic figures. Appendix B contains the full executable R code, and Appendix C reproduces the console output. Formula environment memory addresses in that output are R implementation details, not statistical results. Confidence regions and Bonferroni limits are omitted as instructed.

The archived extended exploration is not necessary to reproduce the main course-level analysis. No observations or unsuccessful candidate models have been removed to improve the reported fit.

References

[1] MH3510 course material. *Modeling linear regression model with R*, Sections 2–7. Practical workflow, model checking, reduced-model tests and prediction.

[2] MH3510 course material. *Chapter 2: Simple linear regression*, Part 4, printed slides 48–59 and 64. Local file: `MH3510_Chapter2 (1).pdf`.

[3] MH3510 course material. *Chapter 4: Multiple linear regression, testing and prediction*, Part I. Local file: `MH3510_Chapter4_Part1.pdf`.

[4] MH3510 course material. *Chapter four: Multiple linear regression (Part II)*, PDF pages 12–31 and 35–36. Local file: `MH3510_CH4part2(ppt).pdf`.

[5] Course project dataset, `aadt.txt`, and assignment instructions. Only the first five columns are used.

A Remaining candidate diagnostics

M1 and M11 appear in the main report. The following figures preserve all other candidate fits. Model specifications and decisions are listed in Section 3.

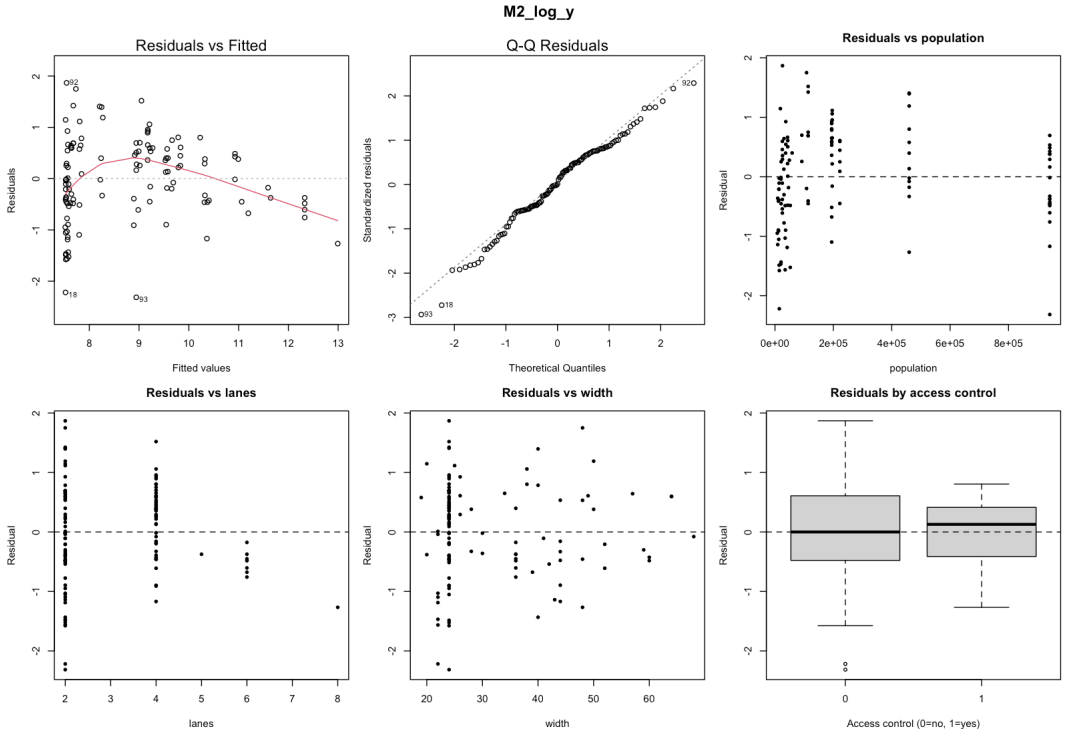


Figure 4: M2: log response with untransformed predictors. Improved Q–Q plot, but a curved residual trend remains.

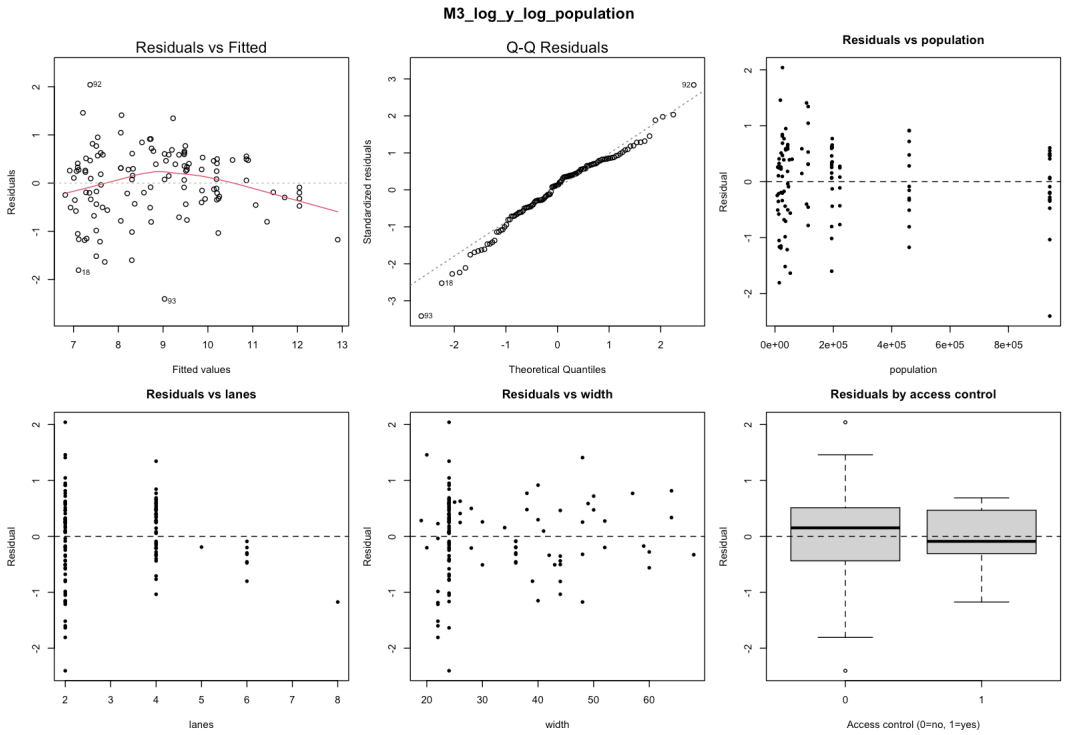


Figure 5: M3: log response and log population. Curvature and changing spread still motivate revision.

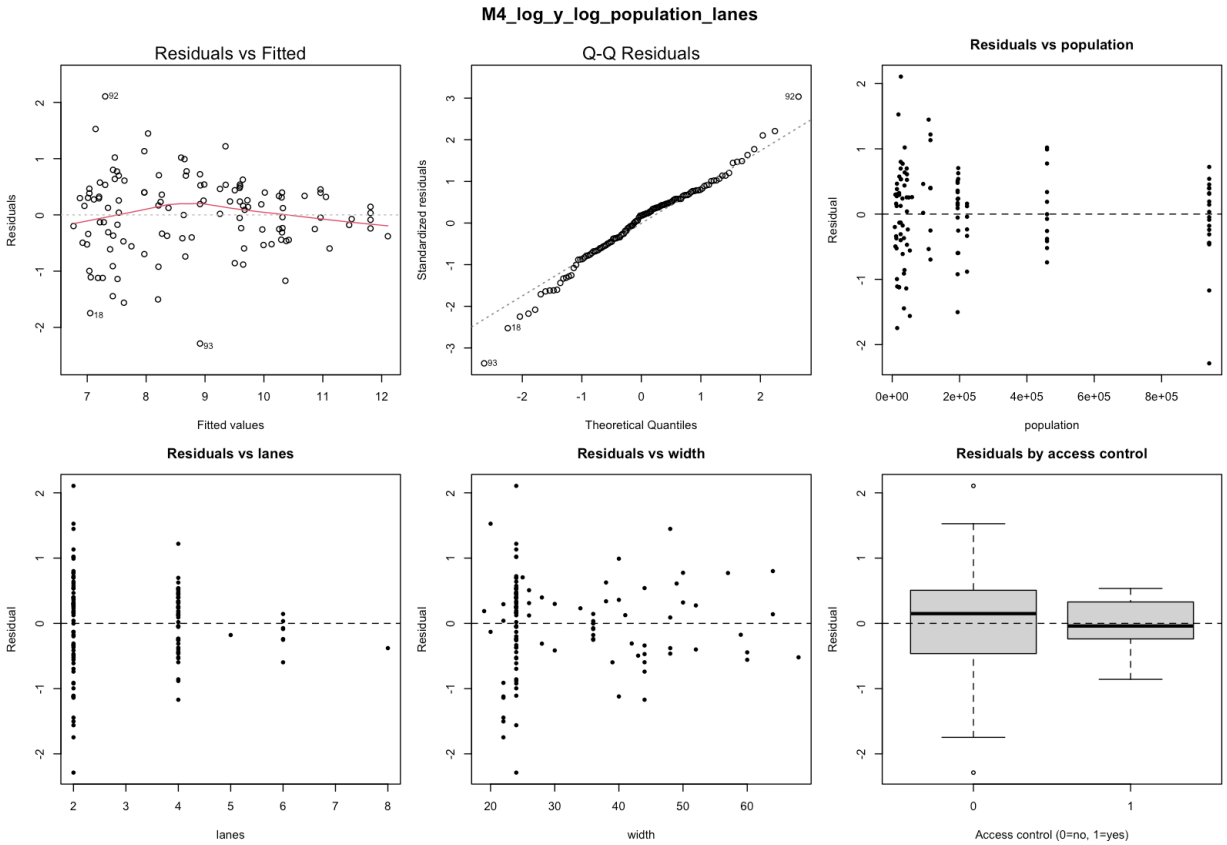


Figure 6: M4: log response, log population and log lanes. The spread still differs across fitted levels.

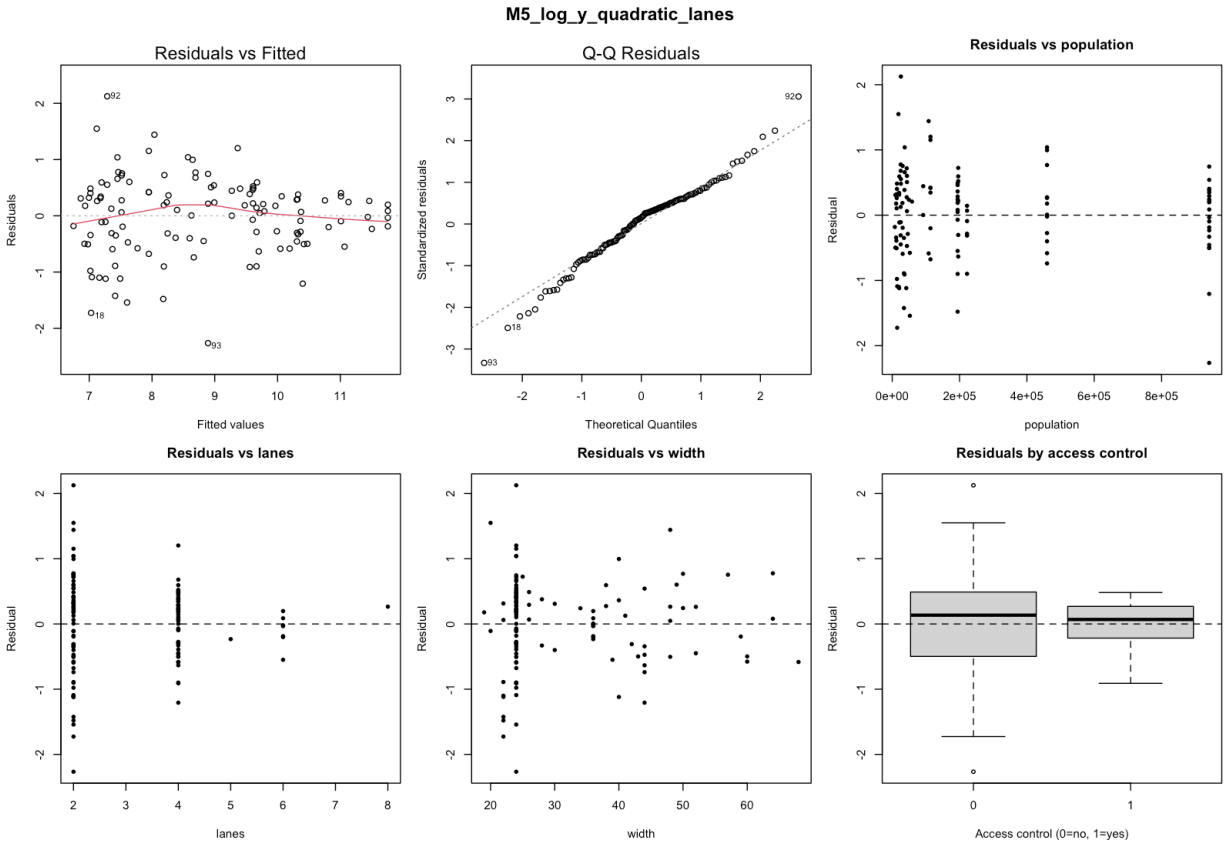


Figure 7: M5: a quadratic lane term improves the log model's mean specification but does not fully resolve uneven spread.

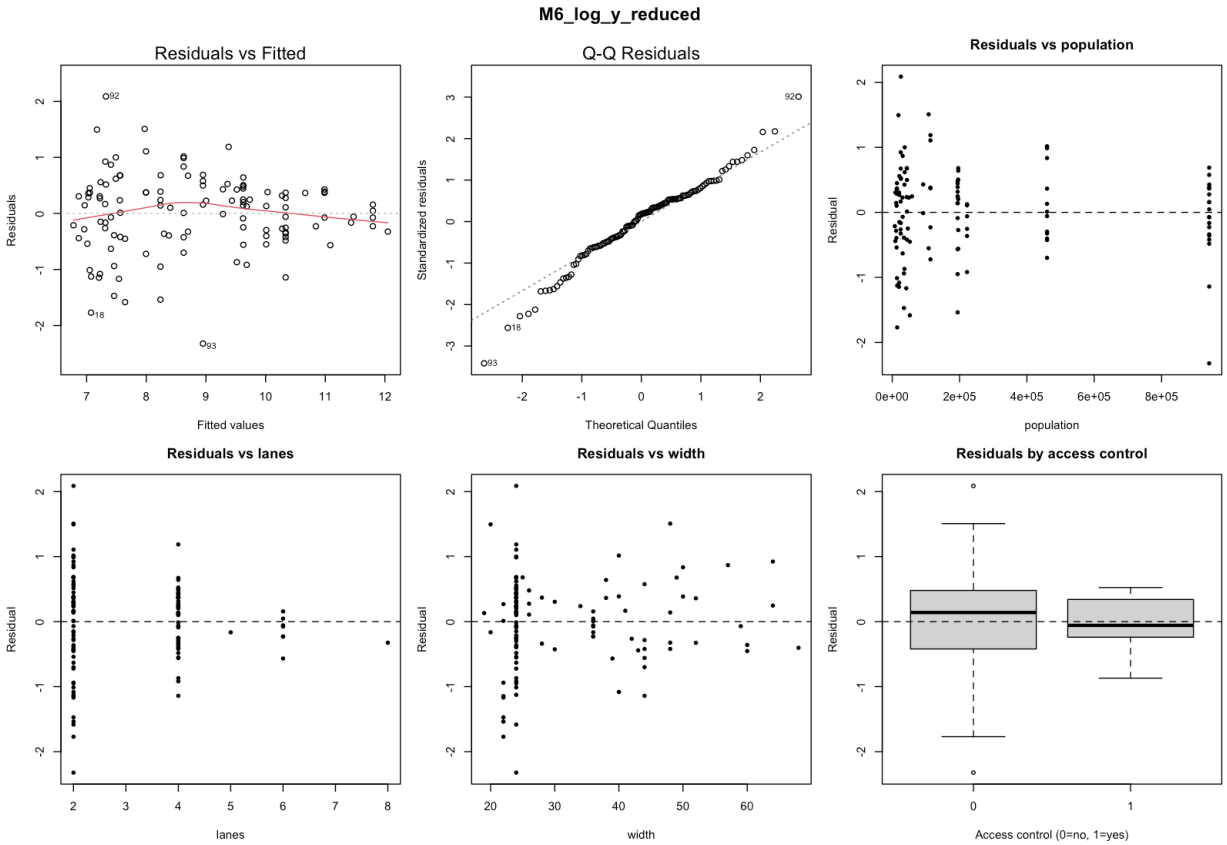


Figure 8: M6: dropping width simplifies the log model but does not repair its variance issue.

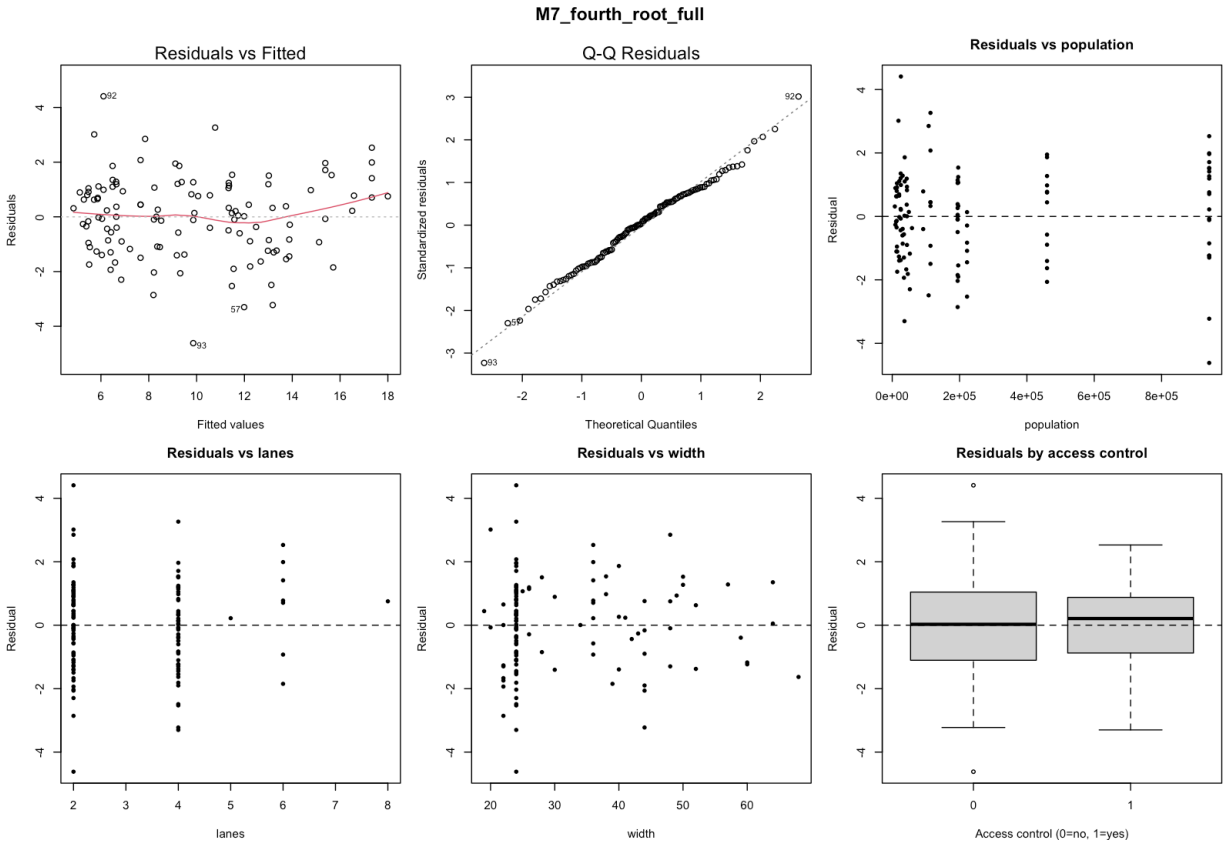


Figure 9: M7: the full fourth-root model is a reasonable transformed candidate with substantially improved diagnostics.

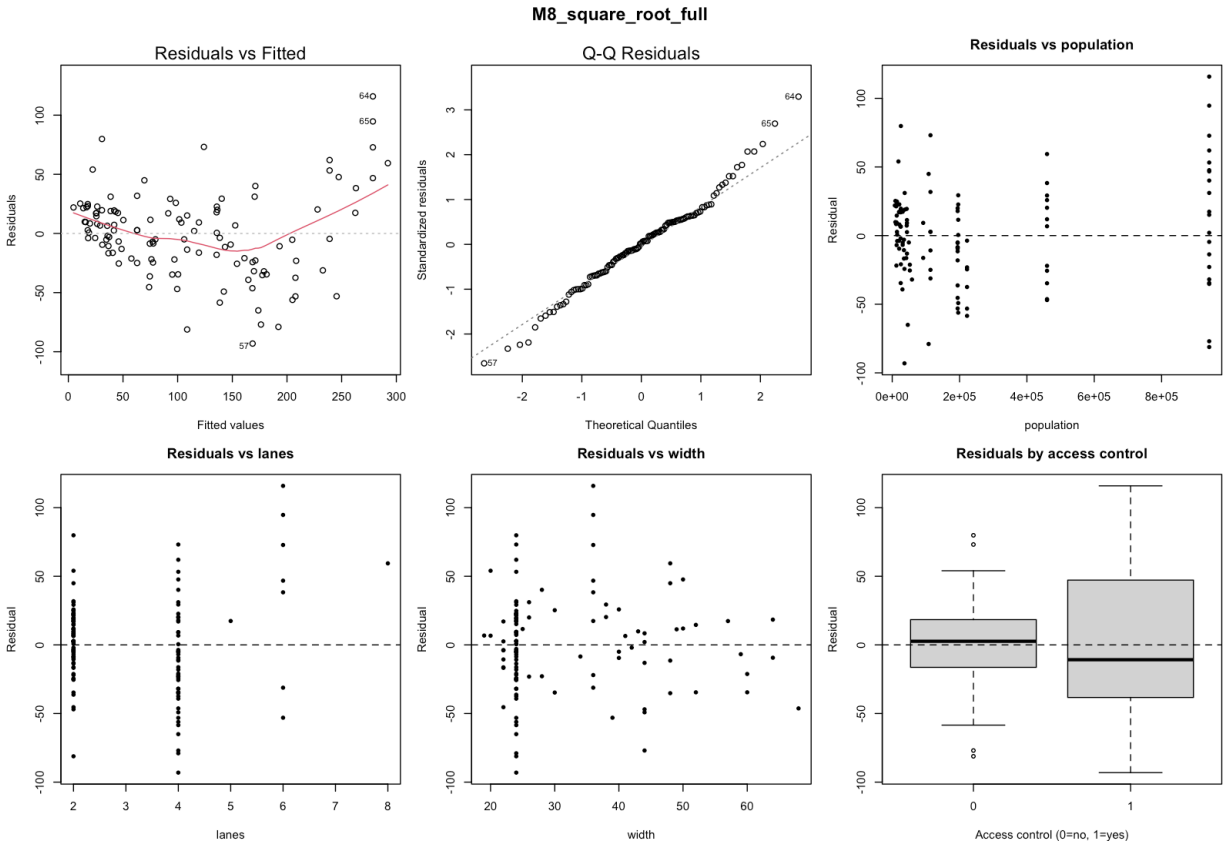


Figure 10: M8: the square-root response model retains curvature and increasing residual spread.

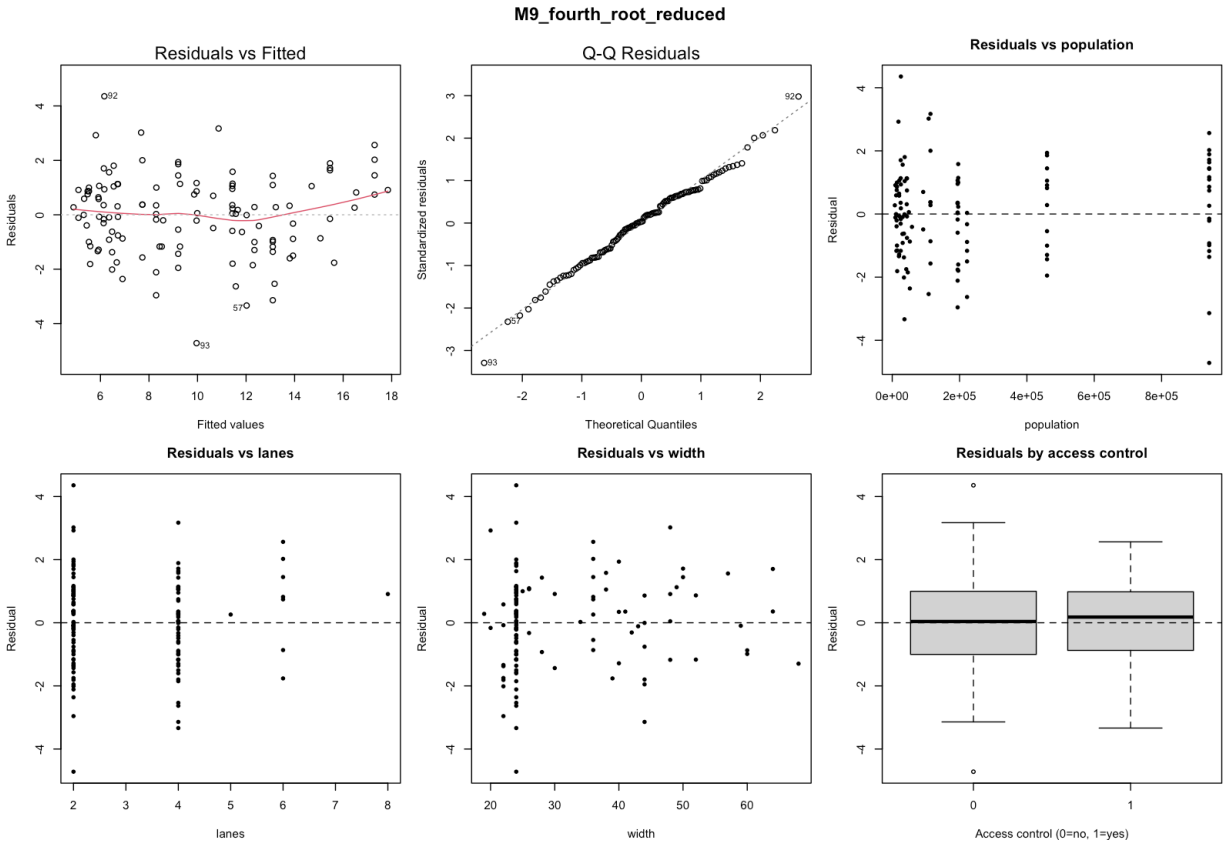


Figure 11: M9: the reduced fourth-root model remains a reasonable alternative to the final specification.

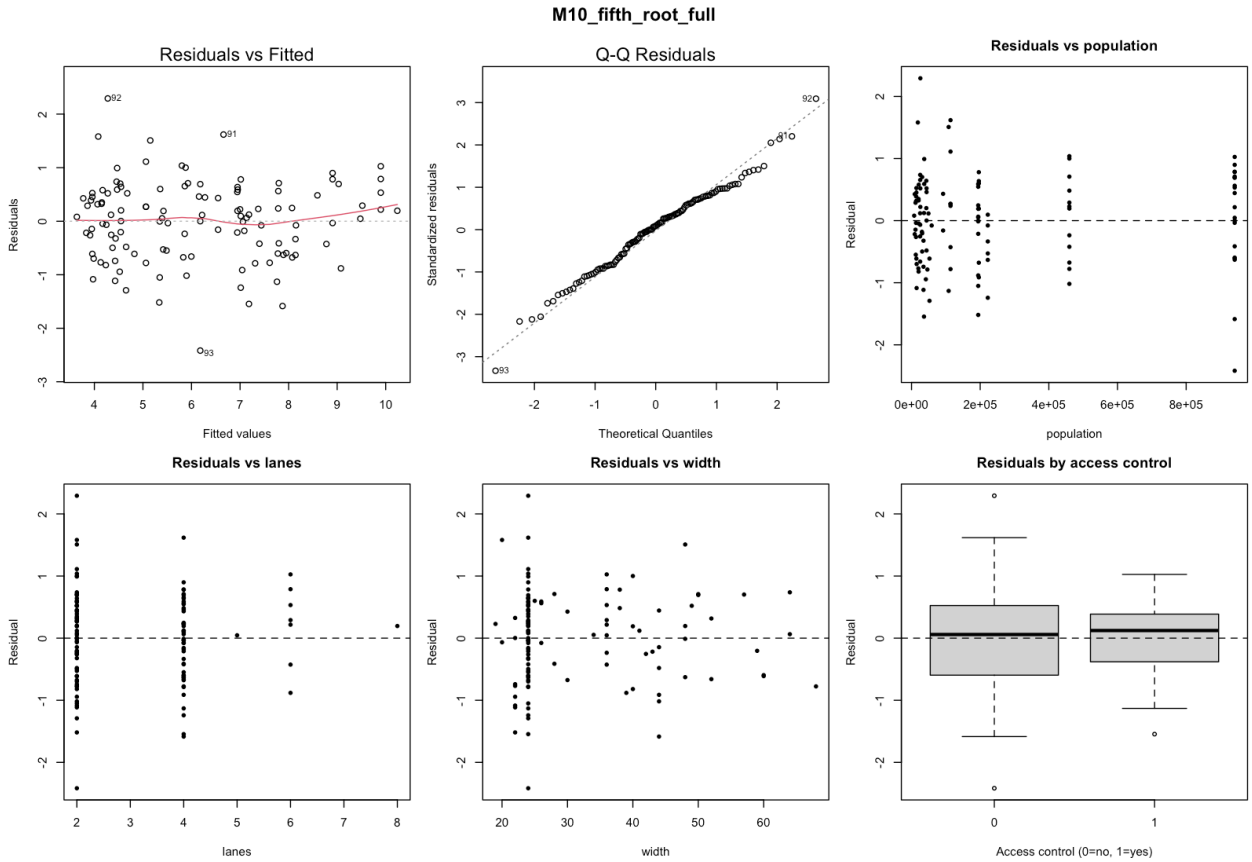


Figure 12: M10: the full fifth-root model. The reduced model M11 omits width after the nested comparison.

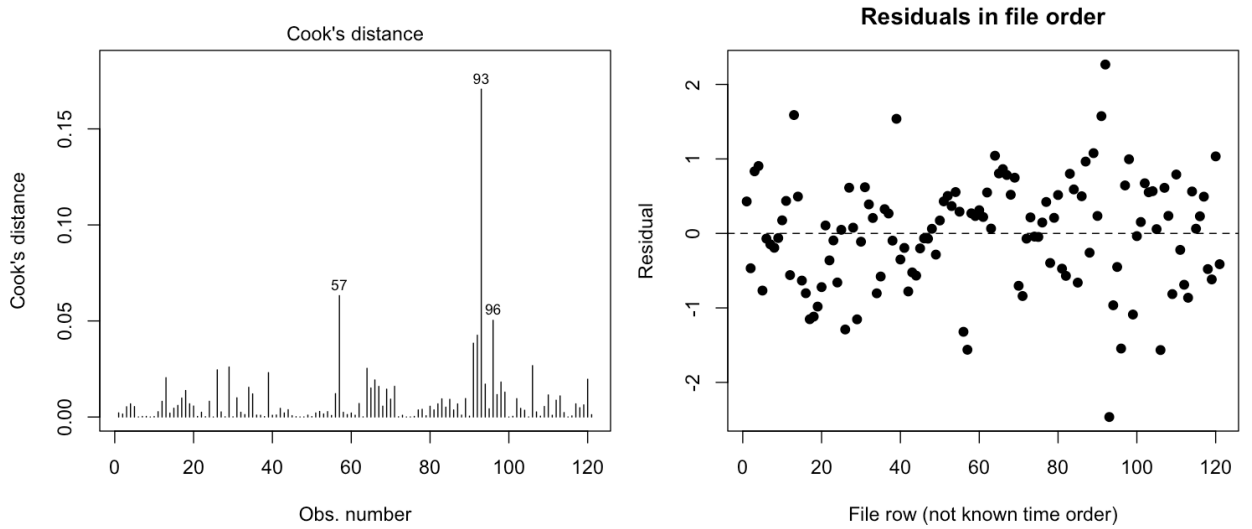


Figure 13: Final-model Cook's distance and residuals in file order. File order is not established to represent time.

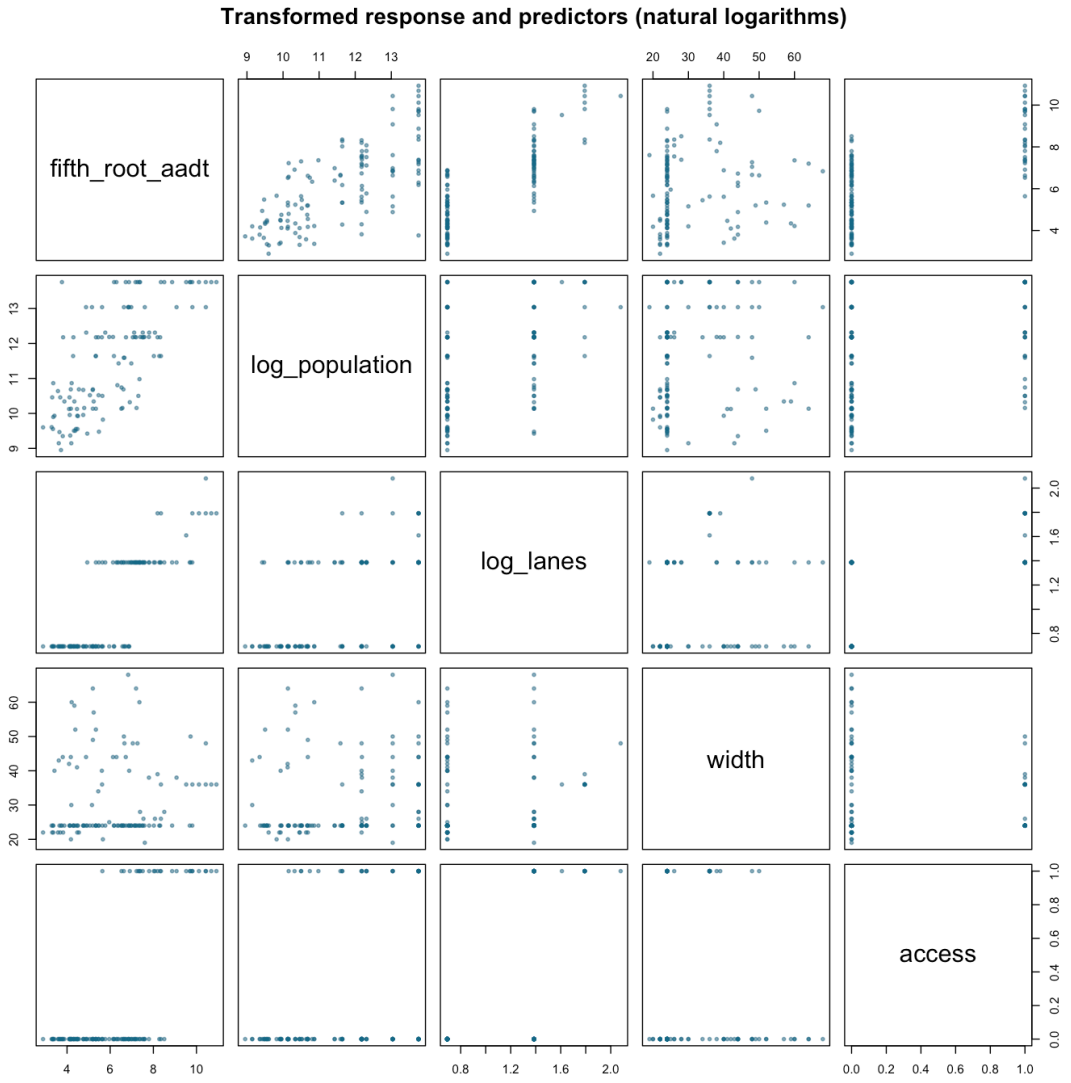


Figure 14: Transformed response and predictors. Width remains displayed to assess the omitted variable.

B Complete R code

The code below is the executable `analysis.R`. Helper functions only format output and save figures. Model fitting and inference use base R.

```r
# MH3510 AADT project - course-aligned analysis, using only base R.
# Run: Rscript analysis.R (with this project folder as the working directory).
# Read analysis_notes.txt for explanations, lecture references and model decisions.
# Results go to coursework_results/. Every candidate model is retained.
# Earlier extended methods are preserved in archive/extended_analysis/.

run_analysis <- function() {
  options(width = 110, digits = 6, useFancyQuotes = FALSE)
  dir.create("coursework_results", showWarnings = FALSE)
  dir.create("coursework_results/plots", showWarnings = FALSE)
  sink("coursework_results/R_output.txt", split = TRUE)
  on.exit(sink(), add = TRUE)

  # These helpers only format output and save plots; they do not fit models.
  section <- function(title) cat("\n\n", title, "\n", strrep("=", nchar(title)), "\n", sep = "")
  save_plot <- function(filename, draw, width = 1600, height = 1100) {
    png(file.path("coursework_results/plots", filename), width, height, res = 140)
    on.exit(dev.off())
    draw()
  }

  section("STEP 1: LOAD AND EXPLORE THE DATA")
  raw <- read.table("aadt.txt", header = FALSE)
  d <- raw[, 1:5]
  names(d) <- c("aadt", "population", "lanes", "width", "control_original")
  stopifnot(!anyNA(d), all(is.finite(as.matrix(d))), all(d$aadt > 0),
            all(d$population > 0), all(d$lanes > 0),
            all(d$control_original %in% c(1, 2)))
  d$access <- ifelse(d$control_original == 1, 1, 0)
  cat("Number of road sections:", nrow(d), "\n")
  cat("access=1: access control; access=0: no access control.\n")
  print(summary(d))
  print(with(d, table(lanes, control_original)))
  print(cor(d[, c("aadt", "population", "lanes", "width", "access")]))
  write.csv(d, "coursework_results/analysis_data.csv", row.names = FALSE)
  save_plot("01_scatterplot_matrix.png", function() {
    pairs(d[, c("aadt", "population", "lanes", "width", "access")],
          pch = 19, cex = 0.6, col = adjustcolor("#176b87", 0.55),
          main = "AADT and the four candidate predictors")
  }, 1400, 1400)

  section("STEP 2: FIT THE INITIAL MULTIPLE REGRESSION")
  M1 <- lm(aadt ~ population + lanes + width + access, data = d)
  print(summary(M1))
  cat("Negative fitted AADT values:", sum(fitted(M1) < 0), "\n")
  cat("A significant overall F-test does not guarantee satisfactory residuals.\n")

  section("STEP 3: CHECK RESIDUALS, TRY TRANSFORMATIONS, AND RECHECK")
  # Chapter 2, slides 51-59: transform x for curvature and y for uneven spread.
  # Each model below is kept, even if it does not improve the diagnostics.
  M2 <- lm(log(aadt) ~ population + lanes + width + access, data = d)
  M3 <- lm(log(aadt) ~ log(population) + lanes + width + access, data = d)
  M4 <- lm(log(aadt) ~ log(population) + log(lanes) + width + access, data = d)
  # Chapter 4 Part II: polynomial terms are still linear regression coefficients.
  M5 <- lm(log(aadt) ~ log(population) + lanes + I(lanes^2) + width + access, data = d)
  M6 <- lm(log(aadt) ~ log(population) + log(lanes) + access, data = d)
  # A fourth root is sqrt(sqrt(y)); a fifth root is y^(1/5).
  M7 <- lm(I(aadt^0.25) ~ log(population) + log(lanes) + width + access, data = d)
  M8 <- lm(sqrt(aadt) ~ log(population) + log(lanes) + width + access, data = d)
```

```r
M9 <- lm(I(aadt^0.25) ~ log(population) + log(lanes) + access, data = d)
M10 <- lm(I(aadt^0.2) ~ log(population) + log(lanes) + width + access, data = d)
M11 <- lm(I(aadt^0.2) ~ log(population) + log(lanes) + access, data = d)
models <- list(M1_raw = M1, M2_log_y = M2, M3_log_y_log_population = M3,
               M4_log_y_log_population_lanes = M4, M5_log_y_quadratic_lanes = M5,
               M6_log_y_reduced = M6, M7_fourth_root_full = M7,
               M8_square_root_full = M8, M9_fourth_root_reduced = M9,
               M10_fifth_root_full = M10, M11_fifth_root_reduced = M11)
response_scales <- c("AADT", rep("log(AADT)", 5), "AADT^0.25", "sqrt(AADT)",
                     "AADT^0.25", "AADT^0.2", "AADT^0.2")
fit_table <- data.frame()
for (i in seq_along(models)) {
  m <- models[[i]]
  id <- names(models)[i]
  cat("\n", id, "\n", sep = "")
  print(formula(m))
  print(summary(m))
  fit_table <- rbind(fit_table, data.frame(
    model = id, response = response_scales[i], number_of_coefficients = length(coef(m)),
    r_squared = summary(m)$r.squared, adjusted_r_squared = summary(m)$adj.r.squared,
    residual_sd = sigma(m)))
  save_plot(paste0(id, "_residuals.png"), function() {
    par(mfrow = c(2, 3), mar = c(4, 4, 3, 1), oma = c(0, 0, 2, 0))
    plot(m, which = 1, sub.caption = "", ask = FALSE)
    plot(m, which = 2, sub.caption = "", ask = FALSE)
    for (v in c("population", "lanes", "width")) {
      plot(d[[v]], residuals(m), xlab = v, ylab = "Residual", pch = 19, cex = 0.6,
           main = paste("Residuals vs", v))
      abline(h = 0, lty = 2)
    }
    boxplot(residuals(m) ~ d$access, xlab = "Access control (0=no, 1=yes)",
            ylab = "Residual", main = "Residuals by access control")
    abline(h = 0, lty = 2)
    mtext(id, outer = TRUE, font = 2)
  })
}
print(fit_table, row.names = FALSE)
write.csv(fit_table, "coursework_results/model_fit_summary.csv", row.names = FALSE)
saveRDS(models, "coursework_results/all_candidate_models.rds")
cat("Compare R-squared, adjusted R-squared and residual SD only on the SAME response scale.\n")
cat("Across different response scales, judge residual patterns and explain the transformation.\
  n")

section("STEP 4: FULL/REDUCED MODEL F-TESTS AND FINAL MODEL")
cat("Log model: test whether road width is needed, H0: beta_width=0.\n")
print(anova(M6, M4))
cat("Log model: test whether the quadratic lane term is needed.\n")
print(anova(M3, M5))
cat("Fourth-root model: test whether road width is needed.\n")
print(anova(M9, M7))
cat("Fifth-root model: test whether road width is needed.\n")
print(anova(M11, M10))
# Never use anova() to compare different transformed responses.
# Fifth-root residuals have a relatively flat trend, much more stable spread
# than the raw/square-root models, and a reasonable Q-Q plot.
# Fourth-root models are also reasonable alternatives, not failed fits.
final_model <- M11
cat("\nSelected model:\n")
print(formula(final_model))
print(summary(final_model))
cat("Ordinary individual coefficient confidence intervals:\n")
print(confint(final_model))
cat("No confidence region or Bonferroni limits are calculated.\n")
cat("Optional curvature check (Chapter 4 Part II polynomial regression):\n")
curved_model <- lm(I(aadt^0.2) ~ log(population) + log(lanes) + access +
```

```
                      I(log(population)^2) + I(log(lanes)^2), data = d)
print(anova(final_model, curved_model))

# Chapter 2 slides 49 and 53: inspect influence and report sensitivity.
# Do not remove a road merely because its residual is large.
influential_row <- which.max(cooks.distance(final_model))
cat("Largest Cook's-distance row:", influential_row, "\n")
print(d[influential_row, ])
sensitivity_model <- lm(formula(final_model), data = d[-influential_row, ])
print(cbind(all_rows = coef(final_model), omit_one_for_sensitivity = coef(sensitivity_model)))
cat("Final model retains ALL road sections. Sensitivity fit is only a check.\n")
write.csv(data.frame(row = seq_len(nrow(d)),
                     standardized_residual = residuals(final_model) / sigma(final_model),
                     cooks_distance = cooks.distance(final_model)),
          "coursework_results/influence_check.csv", row.names = FALSE)
saveRDS(list(curvature_check = curved_model, sensitivity_only = sensitivity_model),
        "coursework_results/additional_checks.rds")
save_plot("02_final_influence_and_order.png", function() {
  par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
  plot(final_model, which = 4, sub.caption = "", ask = FALSE)
  plot(residuals(final_model), xlab = "File row (not known time order)",
       ylab = "Residual", pch = 19, main = "Residuals in file order")
  abline(h = 0, lty = 2)
}, 1500, 650)
# Reference tutorial gives this statistic; order is not documented here.
e <- residuals(final_model)
durbin_watson <- sum(diff(e)^2) / sum(e^2)
cat("Durbin-Watson statistic in FILE order:", durbin_watson, "\n")
cat("No time/spatial order is provided, so no serial-correlation conclusion is drawn.\n")
cat("Independence remains an assumption; a row-order plot cannot establish it.\n")

section("STEP 5: PREDICTION FOR THE REQUESTED ROAD")
new_road <- data.frame(population = 50000, lanes = 3, width = 60, access = 0)
cat("Original control code x4=2 means NO access control, hence access=0.\n")
cat("Width=60 is recorded, but width is excluded from the selected model.\n")
print(new_road)
# Standard predict() intervals, as in the reference tutorial and Chapter 4.
mean_root_interval <- predict(final_model, new_road, interval = "confidence", level = 0.95)
individual_root_interval <- predict(final_model, new_road, interval = "prediction", level =
  0.95)
cat("95% confidence interval for mean FIFTH-ROOT AADT:\n")
print(mean_root_interval)
cat("95% prediction interval for one road, on the FIFTH-ROOT scale:\n")
print(individual_root_interval)
# Fifth power converts the individual prediction interval to AADT units.
# It does NOT turn a mean-root confidence interval into a mean-AADT interval.
point_aadt <- mean_root_interval[1, "fit"]^5
individual_aadt_interval <- individual_root_interval[1, c("lwr", "upr")]^5
cat("Back-transformed fitted value (typical AADT):", point_aadt, "vehicles/day\n")
cat("95% individual-road prediction interval in vehicles/day:\n")
print(individual_aadt_interval)
cat("This point is a typical/approximate median prediction, NOT the arithmetic mean AADT.\n")
cat("A mean-AADT CI requires an extra retransformation method; it is not claimed here.\n")
cat("There are no 3-lane observations; 3 lanes interpolates between 2 and 4 lanes.\n")
cat("Sensitivity-only back-transformed point:", as.numeric(predict(sensitivity_model, new_road)
  )^5, "\n")
cat("Fourth-root alternative's back-transformed point:", as.numeric(predict(M9, new_road))^4, "
  \n")
predictions <- data.frame(
  quantity = c("Mean fifth-root AADT: confidence interval",
               "Typical AADT point with individual-road prediction interval"),
  estimate = c(mean_root_interval[1, "fit"], point_aadt),
  lower_95 = c(mean_root_interval[1, "lwr"], individual_aadt_interval["lwr"]),
  upper_95 = c(mean_root_interval[1, "upr"], individual_aadt_interval["upr"]),
  units = c("(vehicles/day)^0.2", "vehicles/day"))
```

```
  write.csv(predictions, "coursework_results/prediction.csv", row.names = FALSE)
  print(predictions, row.names = FALSE)
  save_plot("03_final_transformed_scatterplots.png", function() {
    pairs(data.frame(fifth_root_aadt = d$aadt^0.2, log_population = log(d$population),
                     log_lanes = log(d$lanes), width = d$width, access = d$access),
          pch = 19, cex = 0.6, col = adjustcolor("#176b87", 0.55),
          main = "Transformed response and predictors (natural logarithms)")
  }, 1400, 1400)
  section("SESSION INFORMATION")
  print(sessionInfo())
  cat("\nRead analysis_notes.txt for the explanation of every candidate model.\n")
}

run_analysis()
```

C Complete R console output

The following is the complete output from the course-level analysis. The model summaries, nested tests, sensitivity results and predictions are included.

```
STEP 1: LOAD AND EXPLORE THE DATA
================================
Number of road sections: 121
access=1: access control; access=0: no access control.
      aadt          population         lanes         width      control_original     access
 Min.   :   201   Min.   :  7716   Min.   :2.0   Min.   :19.0   Min.   :1.00     Min.   :0.000
 1st Qu.:  1769   1st Qu.: 25717   1st Qu.:2.0   1st Qu.:24.0   1st Qu.:2.00     1st Qu.:0.000
 Median :  8666   Median :113571   Median :2.0   Median :24.0   Median :2.00     Median :0.000
 Mean   : 19438   Mean   :263428   Mean   :3.1   Mean   :31.1   Mean   :1.78     Mean   :0.223
 3rd Qu.: 21535   3rd Qu.:459784   3rd Qu.:4.0   3rd Qu.:38.0   3rd Qu.:2.00     3rd Qu.:0.000
 Max.   :155547   Max.   :941411   Max.   :8.0   Max.   :68.0   Max.   :2.00     Max.   :1.000
     control_original
lanes  1  2
    2  0 64
    4 18 30
    5  1  0
    6  7  0
    8  1  0
               aadt population     lanes      width     access
aadt       1.000000   0.620488 0.7644155  0.1218162  0.6854616
population 0.620488   1.000000 0.4168752  0.1664928  0.2758518
lanes      0.764415   0.416875 1.0000000  0.0991997  0.6642230
width      0.121816   0.166493 0.0991997  1.0000000 -0.0472547
access     0.685462   0.275852 0.6642230 -0.0472547  1.0000000


STEP 2: FIT THE INITIAL MULTIPLE REGRESSION
===========================================

Call:
lm(formula = aadt ~ population + lanes + width + access, data = d)

Residuals:
   Min     1Q Median     3Q    Max
-36263  -8501   3493   6018  68317

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.60e+04   5.25e+03   -4.95  2.5e-06 ***
population   3.30e-02   4.71e-03    7.02  1.6e-10 ***
lanes        9.16e+03   1.53e+03    5.98  2.5e-08 ***
width        1.00e+02   1.24e+02    0.81     0.42
access       2.36e+04   4.52e+03    5.22  7.8e-07 ***
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1

Residual standard error: 15300 on 116 degrees of freedom
Multiple R-squared:  0.753,     Adjusted R-squared:  0.744
F-statistic: 88.3 on 4 and 116 DF,  p-value: <2e-16

Negative fitted AADT values: 45
A significant overall F-test does not guarantee satisfactory residuals.


STEP 3: CHECK RESIDUALS, TRY TRANSFORMATIONS, AND RECHECK
========================================================

M1_raw
aadt ~ population + lanes + width + access
<environment: 0x11e419aa8>

Call:
lm(formula = aadt ~ population + lanes + width + access, data = d)

Residuals:
   Min     1Q Median     3Q    Max
-36263  -8501   3493   6018  68317
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.60e+04   5.25e+03   -4.95  2.5e-06 ***
population   3.30e-02   4.71e-03    7.02  1.6e-10 ***
lanes        9.16e+03   1.53e+03    5.98  2.5e-08 ***
width        1.00e+02   1.24e+02    0.81     0.42
access       2.36e+04   4.52e+03    5.22  7.8e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15300 on 116 degrees of freedom
Multiple R-squared:  0.753,     Adjusted R-squared:  0.744
F-statistic: 88.3 on 4 and 116 DF,  p-value: <2e-16


M2_log_y
log(aadt) ~ population + lanes + width + access
<environment: 0x11e419aa8>

Call:
lm(formula = log(aadt) ~ population + lanes + width + access,
    data = d)

Residuals:
    Min      1Q  Median      3Q     Max
-2.3152 -0.4611  0.0096  0.5935  1.8678

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 6.07e+00   2.83e-01   21.49  < 2e-16 ***
population  1.53e-06   2.53e-07    6.03  2.0e-08 ***
lanes       6.85e-01   8.23e-02    8.32  1.9e-13 ***
width       2.50e-03   6.69e-03    0.37    0.709
access      6.14e-01   2.43e-01    2.53    0.013 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.823 on 116 degrees of freedom
Multiple R-squared:  0.741,     Adjusted R-squared:  0.732
F-statistic:   83 on 4 and 116 DF,  p-value: <2e-16


M3_log_y_log_population
log(aadt) ~ log(population) + lanes + width + access
<environment: 0x11e419aa8>

Call:
lm(formula = log(aadt) ~ log(population) + lanes + width + access,
    data = d)

Residuals:
    Min      1Q  Median      3Q     Max
-2.4044 -0.4021  0.0942  0.4787  2.0412

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.47777    0.54683    2.70   0.0079 **
log(population)  0.46164    0.05176    8.92    8e-15 ***
lanes            0.58354    0.07483    7.80    3e-12 ***
width            0.00157    0.00589    0.27   0.7901
access           0.65925    0.21465    3.07   0.0027 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.726 on 116 degrees of freedom
Multiple R-squared:  0.798,     Adjusted R-squared:  0.791
F-statistic:  115 on 4 and 116 DF,  p-value: <2e-16


M4_log_y_log_population_lanes
log(aadt) ~ log(population) + log(lanes) + width + access
<environment: 0x11e419aa8>

Call:
```

```
lm(formula = log(aadt) ~ log(population) + log(lanes) + width +
    access, data = d)

Residuals:
   Min     1Q Median     3Q    Max
-2.287 -0.400  0.124  0.400  2.107

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      1.29875    0.52644    2.47   0.0151 *
log(population)  0.44728    0.05017    8.91  8.2e-15 ***
log(lanes)       1.98900    0.23158    8.59  4.7e-14 ***
width            0.00356    0.00566    0.63   0.5309
access           0.66727    0.20190    3.30   0.0013 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.701 on 116 degrees of freedom
Multiple R-squared:  0.812,     Adjusted R-squared:  0.805
F-statistic:  125 on 4 and 116 DF,  p-value: <2e-16


M5_log_y_quadratic_lanes
log(aadt) ~ log(population) + lanes + I(lanes^2) + width + access
<environment: 0x11e419aa8>

Call:
lm(formula = log(aadt) ~ log(population) + lanes + I(lanes^2) +
    width + access, data = d)

Residuals:
   Min     1Q Median     3Q    Max
-2.265 -0.400  0.104  0.405  2.125

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.50422    0.61869    0.81    0.417
log(population)  0.44620    0.05029    8.87  1.1e-14 ***
lanes            1.25093    0.23202    5.39  3.8e-07 ***
I(lanes^2)      -0.09050    0.02989   -3.03    0.003 **
width            0.00463    0.00578    0.80    0.425
access           0.70112    0.20794    3.37    0.001 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.702 on 115 degrees of freedom
Multiple R-squared:  0.813,     Adjusted R-squared:  0.805
F-statistic:  100 on 5 and 115 DF,  p-value: <2e-16


M6_log_y_reduced
log(aadt) ~ log(population) + log(lanes) + access
<environment: 0x11e419aa8>

Call:
lm(formula = log(aadt) ~ log(population) + log(lanes) + access,
    data = d)

Residuals:
   Min     1Q Median     3Q    Max
-2.322 -0.389  0.125  0.389  2.088

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)       1.3510     0.5185    2.61   0.0104 *
log(population)   0.4518     0.0495    9.12  2.6e-15 ***
log(lanes)        1.9987     0.2305    8.67  2.8e-14 ***
access            0.6519     0.1999    3.26   0.0015 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.699 on 117 degrees of freedom
Multiple R-squared:  0.811,     Adjusted R-squared:  0.806
F-statistic:  168 on 3 and 117 DF,  p-value: <2e-16
```

```
M7_fourth_root_full
I(aadt^0.25) ~ log(population) + log(lanes) + width + access
<environment: 0x11e419aa8>

Call:
lm(formula = I(aadt^0.25) ~ log(population) + log(lanes) + width +
    access, data = d)

Residuals:
   Min     1Q Median     3Q    Max
-4.617 -1.084  0.051  0.991  4.409

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      -7.8466     1.1085   -7.08  1.2e-10 ***
log(population)   1.0427     0.1056    9.87  < 2e-16 ***
log(lanes)        4.5027     0.4876    9.23  1.5e-15 ***
width             0.0101     0.0119    0.85      0.4
access            2.4009     0.4251    5.65  1.2e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.48 on 116 degrees of freedom
Multiple R-squared:  0.857,     Adjusted R-squared:  0.852
F-statistic:  173 on 4 and 116 DF,  p-value: <2e-16


M8_square_root_full
sqrt(aadt) ~ log(population) + log(lanes) + width + access
<environment: 0x11e419aa8>

Call:
lm(formula = sqrt(aadt) ~ log(population) + log(lanes) + width +
    access, data = d)

Residuals:
   Min     1Q Median     3Q    Max
-93.06 -22.00   0.72  19.62 115.83

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     -257.800     27.051   -9.53  3.0e-16 ***
log(population)   21.642      2.578    8.39  1.3e-13 ***
log(lanes)        88.145     11.900    7.41  2.2e-11 ***
width              0.320      0.291    1.10     0.27
access            69.207     10.374    6.67  9.1e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 36 on 116 degrees of freedom
Multiple R-squared:  0.829,     Adjusted R-squared:  0.823
F-statistic:  141 on 4 and 116 DF,  p-value: <2e-16


M9_fourth_root_reduced
I(aadt^0.25) ~ log(population) + log(lanes) + access
<environment: 0x11e419aa8>

Call:
lm(formula = I(aadt^0.25) ~ log(population) + log(lanes) + access,
    data = d)

Residuals:
   Min     1Q Median     3Q    Max
-4.717 -0.999  0.048  0.997  4.355

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)       -7.698      1.093   -7.04  1.4e-10 ***
log(population)    1.056      0.104   10.11  < 2e-16 ***
log(lanes)         4.530      0.486    9.32  8.6e-16 ***
access             2.357      0.422    5.59  1.5e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.47 on 117 degrees of freedom
Multiple R-squared:  0.856,     Adjusted R-squared:  0.852
F-statistic:  231 on 3 and 117 DF,  p-value: <2e-16


M10_fifth_root_full
I(aadt^0.2) ~ log(population) + log(lanes) + width + access
<environment: 0x11e419aa8>

Call:
lm(formula = I(aadt^0.2) ~ log(population) + log(lanes) + width +
    access, data = d)

Residuals:
    Min      1Q  Median      3Q     Max
-2.4167 -0.5484  0.0638  0.5207  2.2926

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    -2.80959    0.56238   -5.00  2.1e-06 ***
log(population)  0.52913    0.05360    9.87  < 2e-16 ***
log(lanes)       2.30420    0.24739    9.31  9.6e-16 ***
width            0.00481    0.00605    0.79     0.43
access           1.12725    0.21568    5.23  7.7e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.749 on 116 degrees of freedom
Multiple R-squared:  0.854,     Adjusted R-squared:  0.849
F-statistic:  169 on 4 and 116 DF,  p-value: <2e-16


M11_fifth_root_reduced
I(aadt^0.2) ~ log(population) + log(lanes) + access
<environment: 0x11e419aa8>

Call:
lm(formula = I(aadt^0.2) ~ log(population) + log(lanes) + access,
    data = d)

Residuals:
   Min     1Q Median     3Q    Max
-2.464 -0.524  0.063  0.503  2.267

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)      -2.739      0.554   -4.94  2.6e-06 ***
log(population)    0.535      0.053   10.10  < 2e-16 ***
log(lanes)         2.317      0.246    9.40  5.6e-16 ***
access             1.107      0.214    5.18  9.5e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.748 on 117 degrees of freedom
Multiple R-squared:  0.853,     Adjusted R-squared:  0.849
F-statistic:  226 on 3 and 117 DF,  p-value: <2e-16

                      model   response number_of_coefficients r_squared adjusted_r_squared residual_sd
                     M1_raw       AADT                      5  0.752741           0.744215 1.52926e+04
                  M2_log_y  log(AADT)                       5  0.741058           0.732129 8.22695e-01
       M3_log_y_log_population  log(AADT)                   5  0.798178           0.791218 7.26310e-01
 M4_log_y_log_population_lanes  log(AADT)                   5  0.811954           0.805470 7.01083e-01
      M5_log_y_quadratic_lanes  log(AADT)                   6  0.813074           0.804947 7.02024e-01
             M6_log_y_reduced  log(AADT)                    4  0.811314           0.806476 6.99268e-01
          M7_fourth_root_full  AADT^0.25                    5  0.856520           0.851573 1.47627e+00
          M8_square_root_full sqrt(AADT)                    5  0.829212           0.823322 3.60244e+01
        M9_fourth_root_reduced  AADT^0.25                   4  0.855626           0.851924 1.47453e+00
          M10_fifth_root_full   AADT^0.2                    5  0.853601           0.848553 7.48941e-01
        M11_fifth_root_reduced   AADT^0.2                   4  0.852804           0.849030 7.47762e-01
Compare R-squared, adjusted R-squared and residual SD only on the SAME response scale.
Across different response scales, judge residual patterns and explain the transformation.


STEP 4: FULL/REDUCED MODEL F-TESTS AND FINAL MODEL
```

```
================================================
Log model: test whether road width is needed, H0: beta_width=0.
Analysis of Variance Table

Model 1: log(aadt) ~ log(population) + log(lanes) + access
Model 2: log(aadt) ~ log(population) + log(lanes) + width + access
  Res.Df   RSS Df Sum of Sq     F Pr(>F)
1    117 57.21
2    116 57.02  1    0.1941 0.395  0.531
Log model: test whether the quadratic lane term is needed.
Analysis of Variance Table

Model 1: log(aadt) ~ log(population) + lanes + width + access
Model 2: log(aadt) ~ log(population) + lanes + I(lanes^2) + width + access
  Res.Df   RSS Df Sum of Sq     F  Pr(>F)
1    116 61.19
2    115 56.68  1     4.517 9.165 0.00305 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Fourth-root model: test whether road width is needed.
Analysis of Variance Table

Model 1: I(aadt^0.25) ~ log(population) + log(lanes) + access
Model 2: I(aadt^0.25) ~ log(population) + log(lanes) + width + access
  Res.Df   RSS Df Sum of Sq     F Pr(>F)
1    117 254.4
2    116 252.8  1     1.576 0.723  0.397
Fifth-root model: test whether road width is needed.
Analysis of Variance Table

Model 1: I(aadt^0.2) ~ log(population) + log(lanes) + access
Model 2: I(aadt^0.2) ~ log(population) + log(lanes) + width + access
  Res.Df   RSS Df Sum of Sq     F Pr(>F)
1    117 65.42
2    116 65.07  1    0.3543 0.632  0.428

Selected model:
I(aadt^0.2) ~ log(population) + log(lanes) + access
<environment: 0x11e419aa8>

Call:
lm(formula = I(aadt^0.2) ~ log(population) + log(lanes) + access,
    data = d)

Residuals:
   Min     1Q Median     3Q    Max
-2.464 -0.524  0.063  0.503  2.267

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)       -2.739      0.554   -4.94  2.6e-06 ***
log(population)    0.535      0.053   10.10  < 2e-16 ***
log(lanes)         2.317      0.246    9.40  5.6e-16 ***
access             1.107      0.214    5.18  9.5e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.748 on 117 degrees of freedom
Multiple R-squared:  0.853,     Adjusted R-squared:  0.849
F-statistic:  226 on 3 and 117 DF,  p-value: <2e-16

Ordinary individual coefficient confidence intervals:
                    2.5 %    97.5 %
(Intercept)     -3.837077 -1.640953
log(population)  0.430284  0.640093
log(lanes)       1.829154  2.805336
access           0.683158  1.529836
No confidence region or Bonferroni limits are calculated.
Optional curvature check (Chapter 4 Part II polynomial regression):
Analysis of Variance Table

Model 1: I(aadt^0.2) ~ log(population) + log(lanes) + access
Model 2: I(aadt^0.2) ~ log(population) + log(lanes) + access + I(log(population)^2) +
    I(log(lanes)^2)
  Res.Df   RSS Df Sum of Sq     F Pr(>F)
```

```
1    117 65.42
2    115 64.59  2    0.8267 0.736  0.481
Largest Cook's-distance row: 93
   aadt population lanes width control_original access
93  756     941411     2    24                2      0
                 all_rows omit_one_for_sensitivity
(Intercept)     -2.739015                -3.02246
log(population)  0.535189                 0.57260
log(lanes)       2.317245                 2.19476
access           1.106497                 1.11335
Final model retains ALL road sections. Sensitivity fit is only a check.
Durbin-Watson statistic in FILE order: 1.77264
No time/spatial order is provided, so no serial-correlation conclusion is drawn.
Independence remains an assumption; a row-order plot cannot establish it.


STEP 5: PREDICTION FOR THE REQUESTED ROAD
=========================================
Original control code x4=2 means NO access control, hence access=0.
Width=60 is recorded, but width is excluded from the selected model.
  population lanes width access
1      50000     3    60      0
95% confidence interval for mean FIFTH-ROOT AADT:
      fit    lwr     upr
1 5.59736 5.4029 5.79183
95% prediction interval for one road, on the FIFTH-ROOT scale:
      fit     lwr     upr
1 5.59736 4.10375 7.09098
Back-transformed fitted value (typical AADT): 5494.36 vehicles/day
95% individual-road prediction interval in vehicles/day:
     lwr      upr
 1163.86 17927.96
This point is a typical/approximate median prediction, NOT the arithmetic mean AADT.
A mean-AADT CI requires an extra retransformation method; it is not claimed here.
There are no 3-lane observations; 3 lanes interpolates between 2 and 4 lanes.
Sensitivity-only back-transformed point: 5429.77
Fourth-root alternative's back-transformed point: 5727.83
                                                 quantity   estimate  lower_95    upper_95
                   Mean fifth-root AADT: confidence interval    5.59736    5.4029     5.79183
 Typical AADT point with individual-road prediction interval 5494.35801 1163.8634 17927.95952
              units
 (vehicles/day)^0.2
       vehicles/day


SESSION INFORMATION
===================
R version 4.6.1 (2026-06-24)
Platform: aarch64-apple-darwin24.6.0
Running under: macOS Sequoia 15.6

Matrix products: default
BLAS:   /opt/homebrew/Cellar/openblas/0.3.34/lib/libopenblasp-r0.3.34.dylib
LAPACK: /opt/homebrew/Cellar/r/4.6.1/lib/R/lib/libRlapack.dylib;  LAPACK version 3.12.1

locale:
[1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8

time zone: Asia/Singapore
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

loaded via a namespace (and not attached):
[1] compiler_4.6.1

Read analysis_notes.txt for the explanation of every candidate model.
```